系統工程

Week 4 課程時間 2026年3月17日

Subject: How To Develop a System Concept

當期課號：**517411,** 永久課號：**MGEM 20024**

學分數：**3**

上課時間：每週二下午 **18:30 to 21:30**

光復校區 陽明石牌校區 跨校課程

開課單位：工管碩

從系統工程的框架來拆解 以色列和美國 為什什麼要攻打伊朗？

Stakeholders, CONOPS, Interfaces, Risks, Trade-Off

上週重點回顧

Structure of Complex Systems

When should Human in systems and When should Human out of a system

Extra (in the homework) How AI could impact a system

3/3 Assignment

Review the different definition of SYSTEM ENGINEERING and what we have shared in the classroom on March 3rd, 2026. Create a YOUR definition of System Engineering from a job you choose.
elaborate the followings:

- The goal of your system
- Who are stakeholders
- What is your Concept of Operations
- What disciplinaries do the system needs?

Homework 1 Review

Systems thinking	Systems engineering	Engineering systems
Focus on process	Focus on whole product	Focus on both process and product
Consideration of issues	Solve complex technical problems	Solve complex interdisciplinary technical, social, and management issues
Evaluation of multiple factors and influences	Develop and test tangible system solutions	Influence policy, processes, and use systems engineering to develop systems solutions
Inclusion of patterns, relationships, and common understanding	Need to meet requirements, measure outcomes, and solve problems	Integrate human and technical domain dynamics and approaches

Progressing…

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Needs

Requirements

Functional Requirements

Architect

Implement

V&V

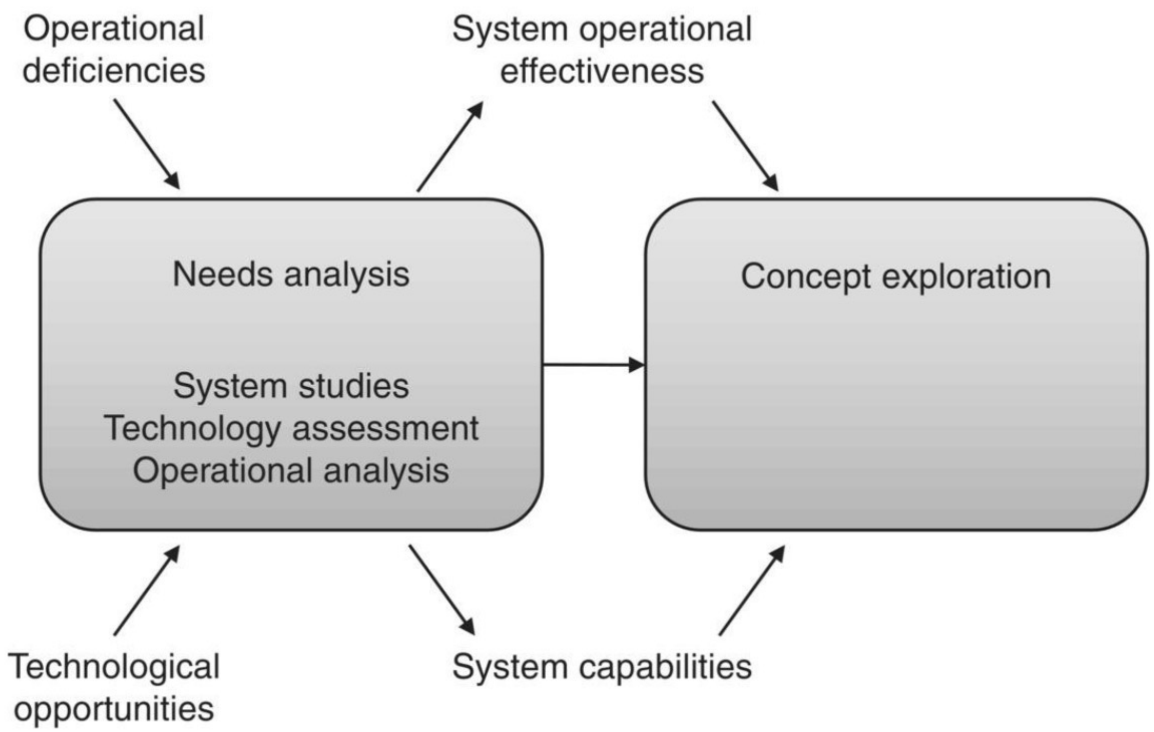
SYSTEMS

Summary of CONCEPT DEVELOPMENT STAGES

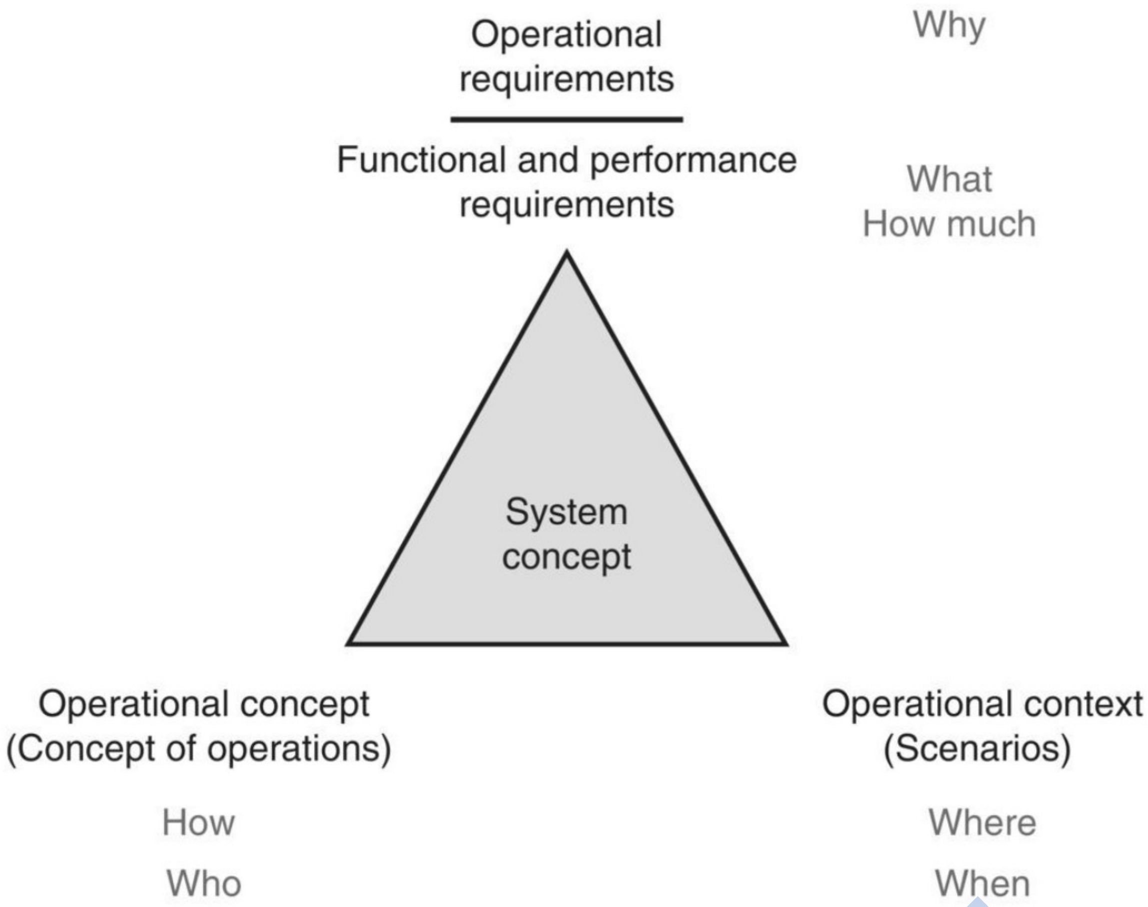
TABLE 5.1. Status of System Materialization at Needs Analysis Phase

Phase Level	*Concept development*			*Engineering development*		
	Needs analysis	**Concept exploration**	**Concept definition**	**Advanced development**	**Engineering design**	**Integration and evaluation**
System	**Define system capabilities and effectiveness**	Identify, explore, and synthesize concepts	Define selected concept with specifications	Validate concept		Test and evaluate
Subsystem		Define requirements and ensure feasibility	Define functional and physical architecture	Validate subsystems		Integrate and test
Component			Allocate functions to components	Define specifications	Design and test	Integrate and test
Subcomponent	*Visualize*			Allocate functions to subcomponents	Design	
Part					Make or buy	

Where NEEDS Come From



What is this chart for?



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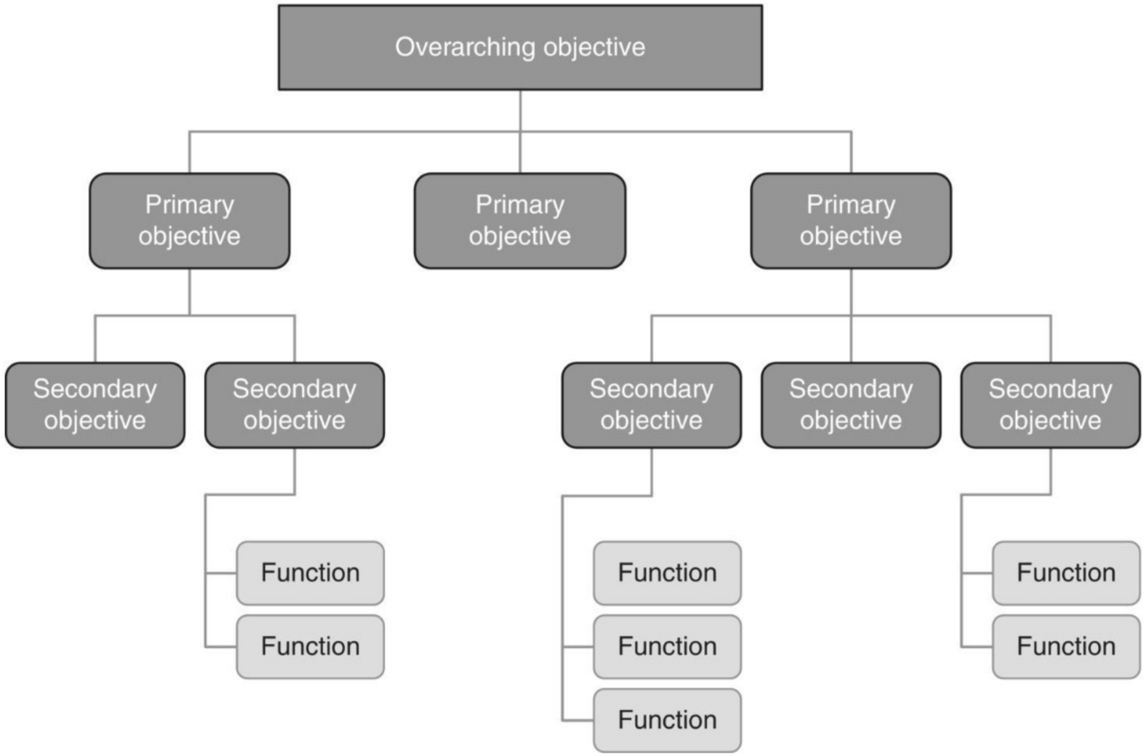
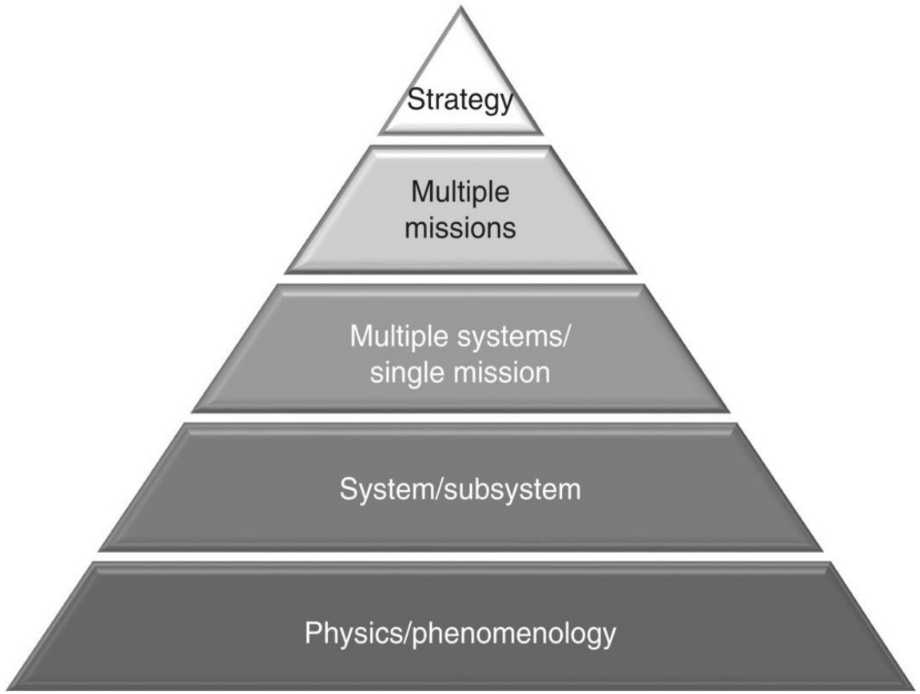


Analysis Pyramid



案例

行動咖啡館

概念採集

移動式服務創新

行動咖啡結合移動式服務，能靈活適應各種場域，提升消費者便利性和新鮮感。

情境模擬互動

透過劇本和情境模擬，探索消費者與品牌的互動，增強體驗感與參與度。

咖啡文化多元發展

行動咖啡促進咖啡文化多元發展，鼓勵創新消費模式和品牌形象塑造。

Five Key Scenario Elements

Mission Objectives

Mission objectives establish the desired outcomes of the scenario and clarify the end goals for all participants.

Friendly Parties

Friendly parties specify allies involved in the scenario, detailing their roles and contributions to achieving objectives.

Threat Actions

Threat actions describe adversaries and their plans, outlining possible challenges and risks within the scenario.

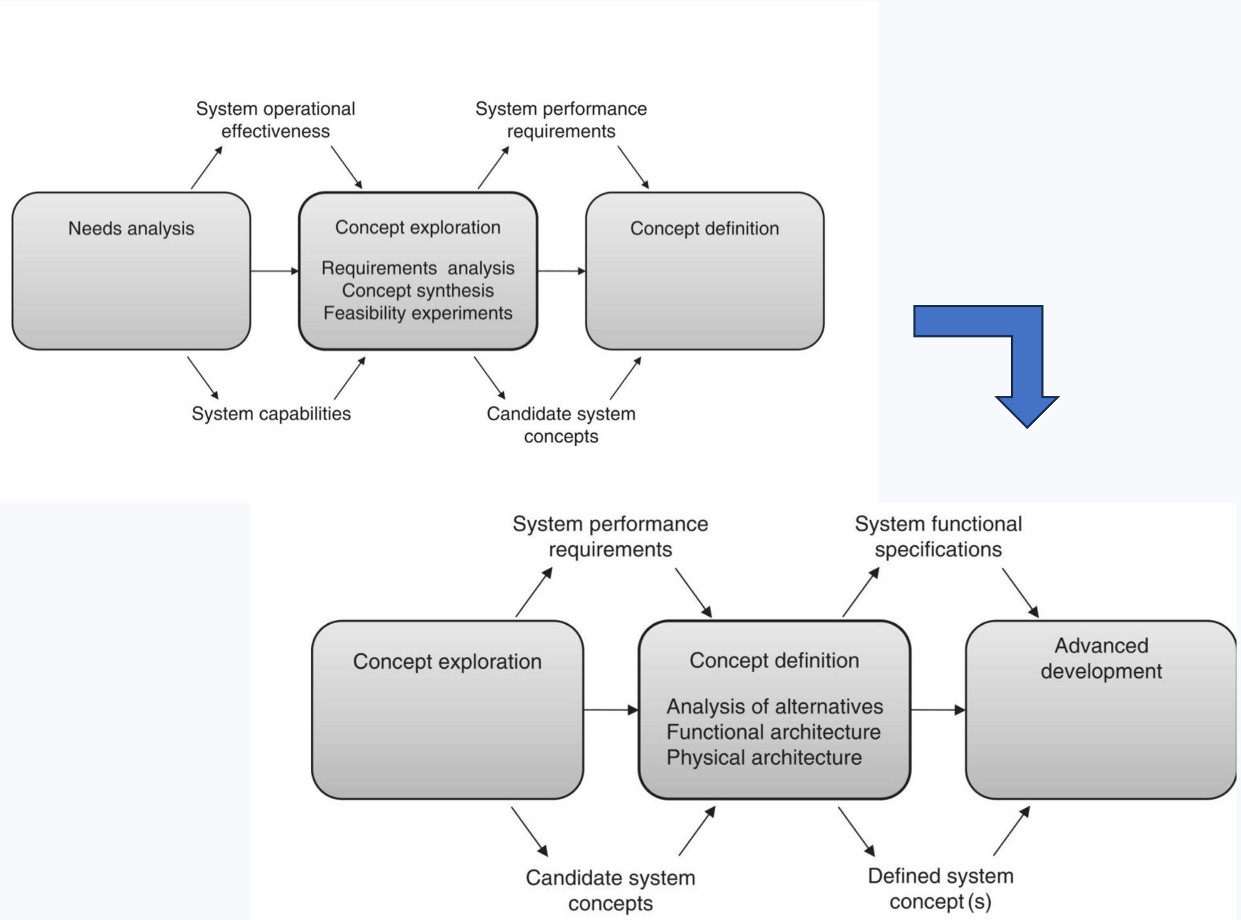
Environment

The environment covers physical, social, or operational surroundings

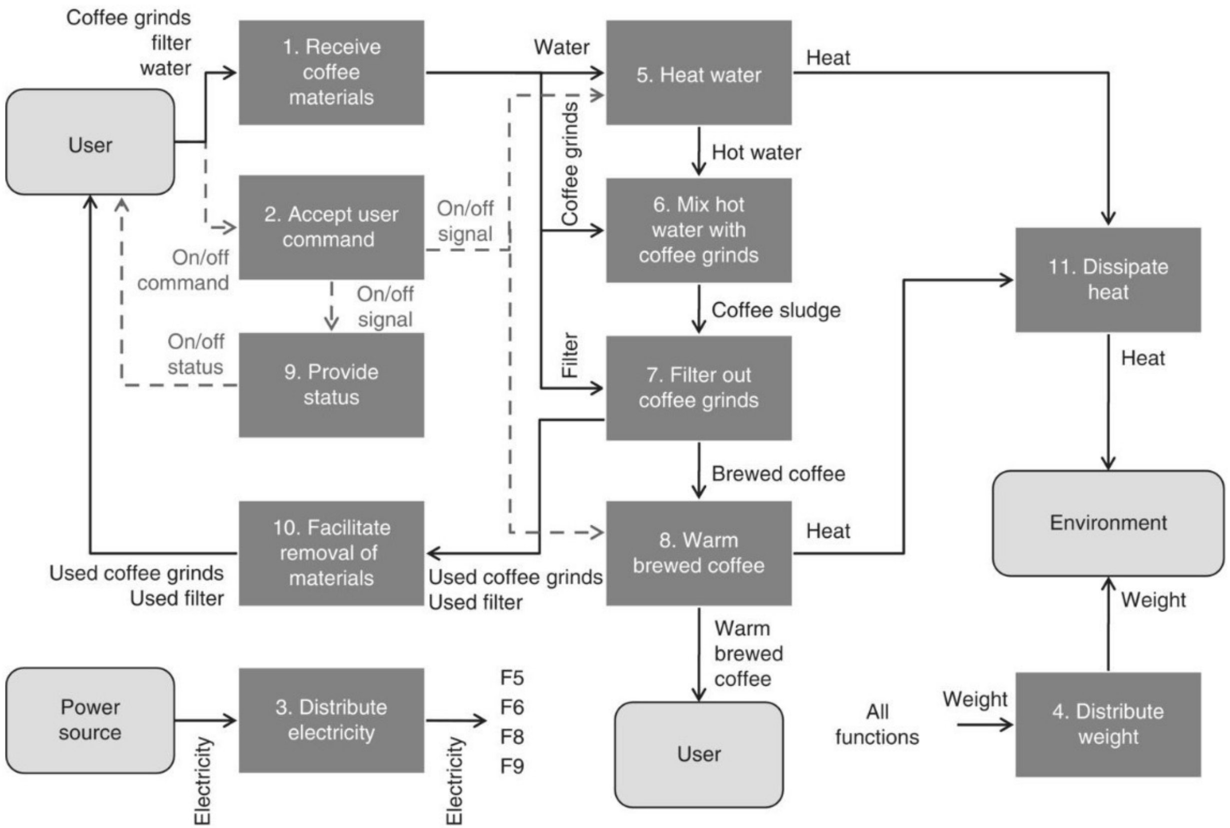
Sequence of Events

the sequence of events traces actions and their cause-and-effect relationships.

Concept Exploration Phase in System Life Cycle



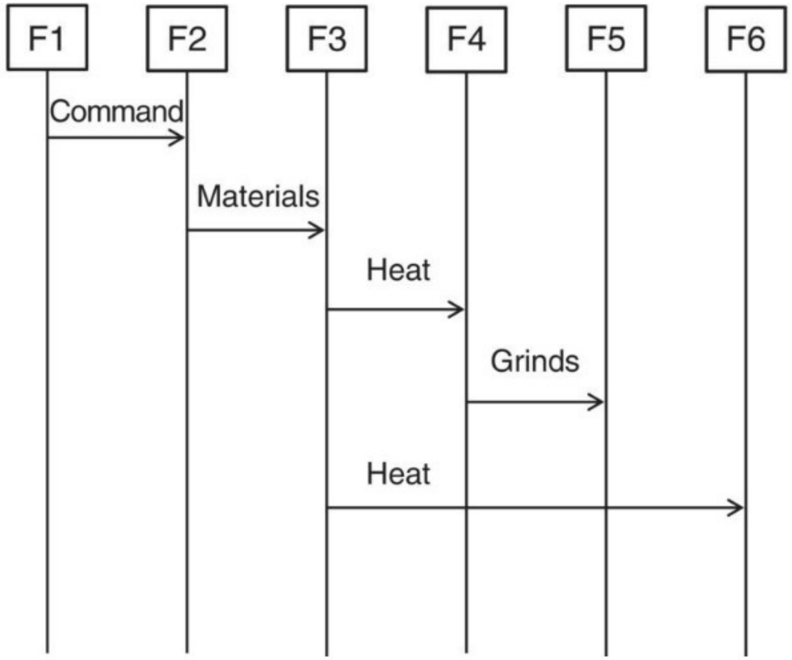
Example of IDEF0 Functional Model Structure

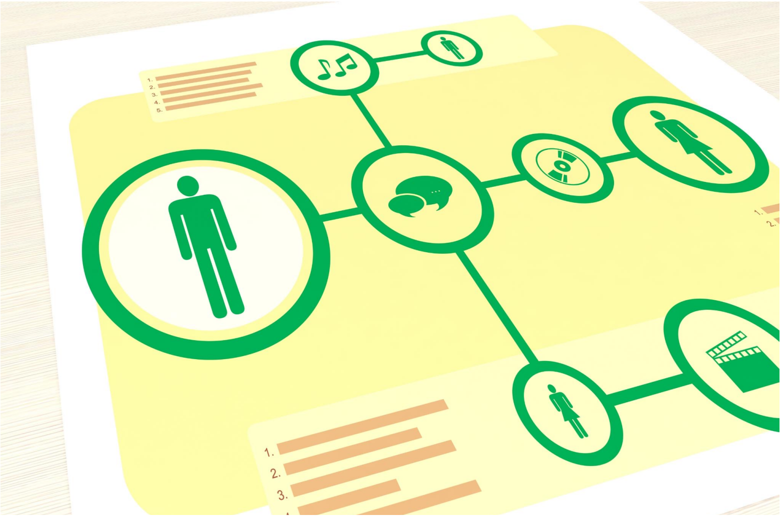


Sequence Diagram / State Transitioning Diagram

System functions:

F1. Accept user command (on/off)
F2. Receive coffee materials
F3. Heat water
F4. Mix hot water with coffee grinds
F5. Filter out coffee grinds
F6. Warm brewed coffee





Common Concept Stage Mistakes

Solving the Wrong Problems

Teams sometimes focus on incorrect or less important issues, leading to wasted effort and missed goals.

Poor Stakeholder Perspective

Failing to consider all user and stakeholder needs results in incomplete or unsatisfactory solutions.

Poor Unclear Requirements

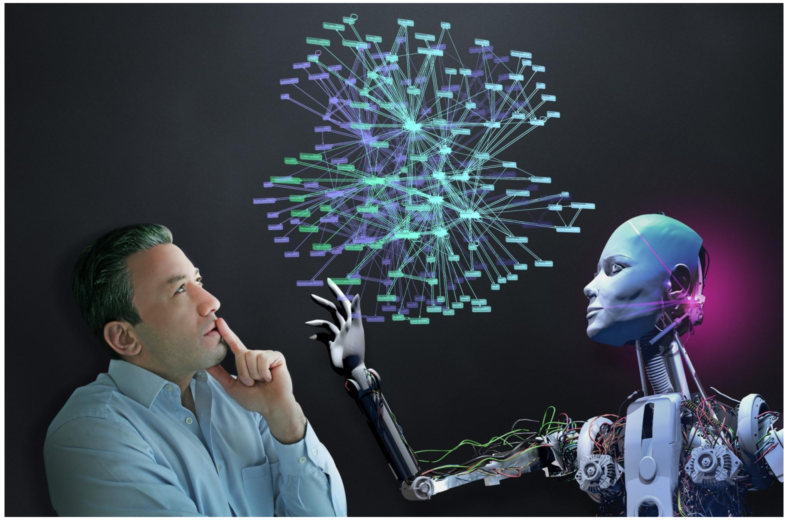
Vague, inconsistent, or untestable requirements cause confusion and hamper project success.

Prematurely narrowing the solution space

Locking into one concept too early before exploring other possibilities. (also vice versa)

Broken Solution Traceability

Disconnects between requirements and functions make it difficult to track project progress and ensure goals are met.



Leveraging AI as a System Engineering Team Member

AI for Meeting Management

AI helps keep and follow up on agendas, ensuring meetings remain organized and on track.

Automated Documentation and Analysis

AI documents meeting minutes, spots logical gaps, and tracks action plans for improved accountability.

AI as Knowledge Base

AI acts as a comprehensive knowledge base, providing quick references for the team's information needs.

AI for Critical Thinking Feedback

AI analyzes, critiques, and suggests optimizations for your team's ideas and decisions, boosting outcomes.

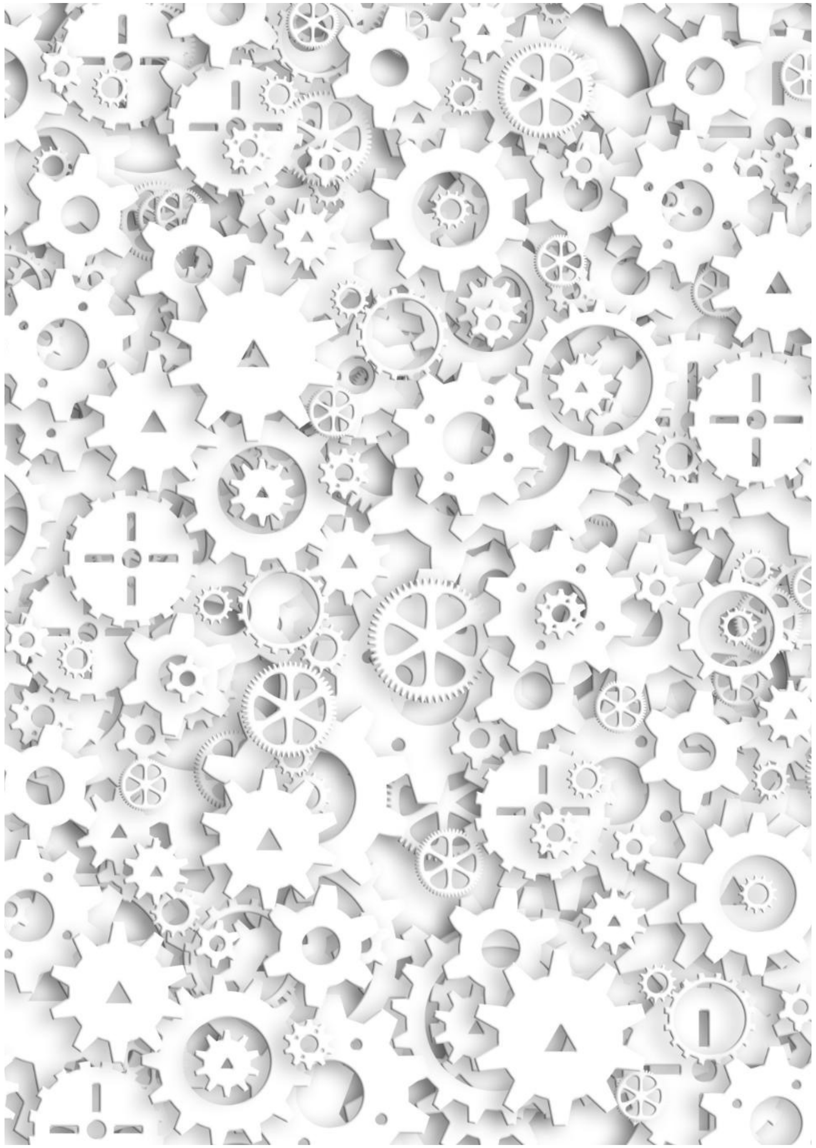
AI as your Project Management Assitant

Naming, coding(numbering) scheme, Version control, Configuration Management, Baseline control

Objectives of This System Engineering Class

透過熟悉系統工程的各種 發散和收斂 的方法，工具，與模型，能夠應用在一個大型系統的各個生命週期上，並且能用專業的系統工程語言，與人和ＡＩ團隊清楚的溝通。

（系統上線之前的思考，系統設計，系統開發，系統測試與驗證，以及系統上線之後的運行狀態，維修，更新升級，以及退役的各個層面）



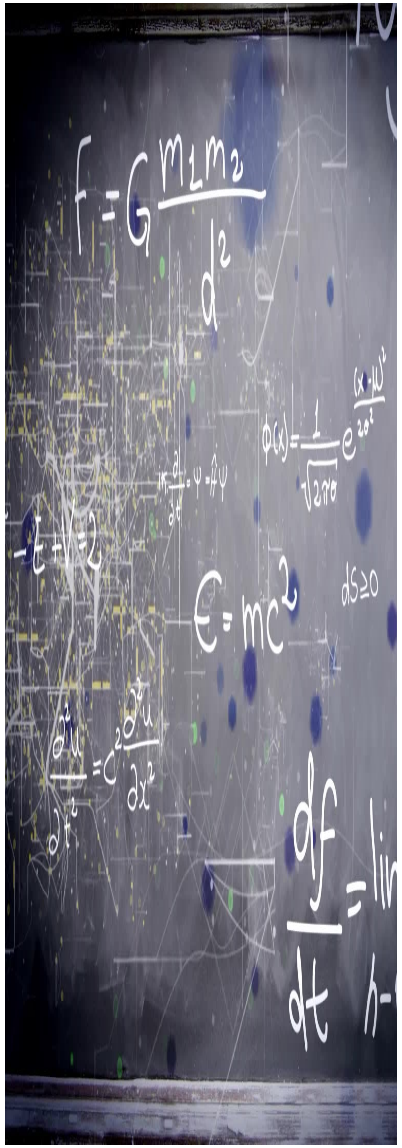
Homework 3

- Reading chapter 5-7
  Key: To be familiar how to use each chart, table and model illustrated in the chapters.

  Decide the subject of your term project (by group)

  Decide the which AI platform to use and why (by group)

  Use AI as +1 team member to summarize your meeting notes and suggest/review your decision of term project and which LLM to use (deliverable : Submit your meeting notes to E3)



End of Week 4

- Homework 3
- Schedule Time to meet with me before the end of this week

- 下週課題：Concept Development Stage (2)
  Evaluation, System Architect,
  (you can preview the chapter 8, 9)

- Coming Homework & Arrangement
  - 3/24,
    Develop a traditional NEEDS analysis by group
  - 3/31,
    Develop a REQUIREMENTS Analysis by group
  - 4/7,
    Develop a FUNTIONAL Analysis & SYSTEM ARCHITECTURE
  - 4/14, Mid-Term
    One hour exam.
    Group presentation of your needs, requirement and functional analysis with your system architecture, and project plan next step

Case Study: After Mid-Term P&G Become a System

A systems-engineering view of how Procter & Gamble evolved into a durable CPG platform

Core argument

P&G did not win only by making better products. It won because it coordinated multiple subsystems—science, packaging, manufacturing, retail, media, and logistics—into one reinforcing operating architecture.

1 System framework

Map P&G as a complex system: nodes, interfaces, feedback loops, and control layers.

2 Revenue timeline

Read annual net sales as evidence of a scalable, resilient CPG operating system.

3 Teaching notes

A classroom-ready way to connect business history, operations, and systems thinking.

P&G is best understood as a system-of-systems, not just a portfolio of brands.